Integral formulas for quaternion valued forms, and their applications to Yang-Mills theory

Edward B. Baker III

University of Maryland, College Park

February 14, 2014

Abstract

We develop an integral formula that relates the inner product of two holomorphic forms as a double integral over the boundary. This result is then generalized to the quaternions, which requires new mathematics that is closely related to Euclidean Yang-Mills theory, and this relationship is explored further.

1 Introduction

The quaternions are an extension of the complex numbers, satisfying the multiplication rules

$$\mathrm{e}_i^2 = -1, \quad [\,\mathrm{e}_i\,,\mathrm{e}_j\,] = 2\,\epsilon_{ijk}\,\mathrm{e}_k \quad \forall\,\mathrm{i,j,k} \in \{1,2,3\}.$$

The quaternions appear naturally in many physical systems, being closely related to the Lie algebra $\mathfrak{su}(2)$ [1][2]. In particular, quaternions are a useful tool in the study of SU(2) Yang Mills theory, and were important for the classification of instantons [3]. There has also been recent interest in hypercomplex geometry (and its variants) in both the mathematics and physics communities [4][5], in addition to the related twistor programme initiated by Roger Penrose [6][7].

In the literature, there has been some emphasis on extending the results of complex analysis and algebraic geometry to the quaternions. The earliest attempt to do this was by Fueter [8][9], related to work by Hamilton, who defines a quaternion regular function by

$$\partial_{\bar{q}} f \equiv \frac{1}{2}(\partial_0 + i\partial_1 + j\partial_2 + k\partial_3)f = 0,$$

which is called the Cauchy-Riemann-Fueter (CRF) equation. This approach has been extended in a number of directions, notably by Dominic Widdows in his thesis [10], related to work by Dominic Joyce [11]. In that work, the CRF equation is extended to the exterior algebra on an arbitrary hypercomplex manifold, and interpreted as a projector onto an invariant subspace of the Lie algebra defined by the generators

$$\mathcal{I}(\omega) = I(\omega) - \omega i, \;\; \mathcal{J}(\omega) = J(\omega) - \omega j, \;\; \mathcal{K}(\omega) = K(\omega) - \omega k,$$

where I, J, and K are the three complex structures associated to the hypercomplex manifold, satisfying $IJ = K$. A different approach was also developed by Cullen [12] [13], which is more closely related to the expansion of a complex analytic function in its power series.

Although the quaternions are naturally related to SU(2) Yang-Mills theory, many constructions formulated in the gauge group SU(2) can be extended to arbitrary Lie groups, such as the ADHM construction [3]. With regards to Yang-Mills theory, there is some motivation to further study the quaternion formulation, due to a Cauchy-type integral formula for quaternion valued functions, as derived by Fueter [8][9]. This could extend results on the holographic principle [15], for which there has been much focus on N=4 Supersymmetric Yang-Mills theory, where the ADS-CFT correspondence can be used to relate bulk and boundary observables [14]. In addition, there has been much recent interest in a newly observed symmetry of Yang-Mills theories, called dual conformal symmetry, that is not evident in the Lagrangian [16][17].

In this paper, we will attempt to derive a holographic integral formula relating the action of a classical Euclidean Yang-Mills field to a double integral over the boundary

$$\int_\Omega \mathrm{Tr}\Big(F(p)\wedge\star F(p)\Big) = \frac{1}{2\pi^2}\int_{\partial\Omega}\int_{\partial\Omega}\mathrm{Tr}\Big(F^+(p)\wedge K(p,q)\wedge F^+(q) - F^-(p)\wedge K(p,q)\wedge F^-(q)\Big), \quad (1.1)$$

where p and q are two quaternions that can be associated to points in $\mathbb{R}^4$, and K(p,q) is an integral kernel defined by

$$K(p,q) = 4\,\mathrm{Re}\left(\frac{\mathrm{d}p\otimes\mathrm{d}\bar{q}}{|p-q|^2}\right) = \sum_\mu \frac{\mathrm{d}x_\mu\otimes\mathrm{d}y_\mu}{|x-y|^2}.$$

This integral formula could suggest a possible semiclassical interpretation of holography for Yang-Mills theory, assuming that the result cannot be generalized to arbitrary fields.

We begin in section 2 with the motivation for this formula, which comes from a calculation in complex analysis. We then generalize to multiple complex dimensions. Section 3 then develops some of the algebra of quaternion valued forms, which is interesting in its own right, and suggests the possibility of new mathematics. In section 4, a number of integral formulas are proved using quaternions, with progressively more physical relevance. Finally, in section 5, we relate this to Yang Mills theory.

2 Complex integral formulas

To begin, we will consider the integral

$$I_1(f,g) = \int_\Omega f(z)g(\bar{z})\,dz\wedge d\bar{z},$$

defined on a closed, compact domain Ω. We first use the Cauchy integral formula for both f(z) and $g(\bar{z})$ to write

$$I_1(f,g) = -\frac{1}{4\pi^2}\int_\Omega\left(\oint_{\partial\Omega}\frac{f(z_1)}{z_1-z}dz_1\oint_{\partial\Omega}\frac{g(\bar{z}_2)}{\bar{z}_2-\bar{z}}d\bar{z}_2\right)dz\wedge d\bar{z}.$$

Assuming this integral does not diverge, we can interchange the loop integrals and the integral over z to obtain

$$I_1(f,g) = -\frac{1}{4\pi^2}\oint_{\partial\Omega}\oint_{\partial\Omega} f(z_1)K(z_1,\bar{z}_2)g(\bar{z}_2)dz_1d\bar{z}_2, \quad (2.1)$$

where we have defined $K(z_1, \bar{z}_2) = \int_\Omega (z - z_1)^{-1} (\bar{z} - \bar{z}_2)^{-1}\, dz \wedge d\bar{z}$. Using the distributional equation $\frac{\partial}{\partial \bar{z}} \frac{1}{z} = \pi \delta^2(z)$, we find that $K(z_1, \bar{z}_2)$ satisfies the differential equation

$$\frac{\partial}{\partial \bar{z}_1} K(z_1, \bar{z}_2) = \frac{\pi}{\bar{z}_1 - \bar{z}_2}, \tag{2.2}$$

with a similar equation for z_2. We therefore find that

$$K(z_1, \bar{z}_2) = \pi \log(|z_1 - z_2|)$$

which is a Green's function for the Laplacian.

We can generalize this formula to higher complex dimension, and then to the quaternions, which will be physically relevant. For the case of n complex variables with $n > 1$, we define a holomorphic form $\omega \in \Omega^{(n,0)}$ and an anti-holomorphic form $\nu \in \Omega^{(0,n)}$, where $\Omega^{(p,q)}$ denotes the space of complex forms of type (p,q). We will derive a formula similar to (2.1) for the integral

$$I_n(\omega, \nu) = \int_\Omega \omega \wedge \nu,$$

in some bounded region $\Omega \in \mathbb{C}^n$. Following the derivation in the complex case, we find an analogue of equation (2.1), with $K(z, w) \in \Omega_1^{(0,n-1)} \otimes \Omega_2^{(n-1,0)}$,

$$I_n(\omega, \nu) = C_n \int_{\partial\Omega} \int_{\partial\Omega} \omega(z) \wedge K_n(z, w) \wedge \nu(w). \tag{2.3}$$

Here K(z, w) must satisfy an equation for the Dolbeault operator, similar to equation (2.2). This solution is related to the Bochner-Martinelli kernel, and we can use the Bochner-Martinelli formula to show,

$$K_n(z, w) = C_n' \sum_{j=1}^{n} |z - w|^{2-2n} \bigwedge_{l \neq j} \mathrm{d}\bar{z}_l \otimes \bigwedge_{m \neq j} \mathrm{d}w_m. \tag{2.4}$$

We can also derive this result in the reverse direction. Starting with equation (2.3), I_n can be rewritten using Stoke's theorem as

$$I_n(\omega, \nu) = C_n \int_\Omega \int_\Omega (\partial_z + \bar{\partial}_z)(\partial_w + \bar{\partial}_w)(\omega(z) \wedge K_n(z, w) \wedge \nu(w)).$$

Because $\omega \in \Omega^{(n,0)}$ and $\bar{\partial}_z \omega = 0$, we have

$$(\partial_z + \bar{\partial}_z)\,(\omega(z) K_n(z, w)) = (-1)^n \omega \wedge \bar{\partial}_z K_n(z, w),$$

and a conjugate result for ν. From this, we see that if K(z,w) satisfies the equation

$$C_n\, \bar{\partial}_z \partial_w K_n(z, w) = \delta^{2n}(z - w) \bigwedge_l \mathrm{d}z_l \otimes \bigwedge_m \mathrm{d}\bar{w}_m + \dots, \tag{2.5}$$

then the result follows immediately, assuming that the extra terms are not of type $\Omega_1^{(n,0)} \otimes \Omega_2^{(0,n)}$, which is true for the kernel (2.4). The extra terms will be considered more explicitly in the quaternionic case. This also explains why $K_n(z, w)$ contains a Green's function for the Laplacian, because if we assume that K_n is only a function of $z' = z - w$, then the left side of equation (2.5) becomes $-C_n \Delta K_n(z - w)$.

3 Quaternion differentials and forms

The generalization to the quaternions is not straightforward, and requires new mathematics that is surprisingly powerful. It is also directly applicable to Euclidean SU(2) Yang-Mills theory in four-dimensions. As motivation for our approach, we consider the formula for the complex differential

$$\mathrm{d}f = \partial_z f \mathrm{d}z + \partial_{\bar{z}} f \mathrm{d}\bar{z}.$$

This formula immediately suggests the splitting between holomorphic and anti-holomorphic functions that is so important in the study of complex analysis. There are a few ways of extending this formula to the quaternions, but one that is both useful and suggestive is

$$\mathrm{d}f = \mathrm{d}\bar{q}\ \partial_{\bar{q}} f + \sum_i (\mathrm{e}_i \mathrm{d}\bar{q}\, \bar{\mathrm{e}}_i)(\mathrm{e}_i \partial_{\bar{q}} \bar{\mathrm{e}}_i\, f). \tag{3.1}$$

From this formula, we see that the CRF operator appears naturally in the expression for the quaternion differential. We define the vector fields and one forms appearing in equation (3.1) as[1]

$$\bar{\not\partial}_\mu \equiv \mathrm{e}_\mu\, \partial_{\bar{q}}\, \bar{\mathrm{e}}_\mu, \quad \bar{\chi}_\mu \equiv \mathrm{e}_\mu\, \mathrm{d}\bar{q}\ \bar{\mathrm{e}}_\mu. \tag{3.2}$$

These operators satisfy the relations

$$\chi_\mu\, (\not\partial_\nu) = \delta_{\mu\nu}, \quad \sum_\mu \not\partial_\mu \not\partial_\mu = \Box, \quad 4\, \bar{\not\partial}_\mu \not\partial_\mu = \Delta,$$

where $\Box$ represents the D'Alembert operator, and the first operation represents a contraction of the one form χ_μ with the vector field $\not\partial_\mu$. We will also refer to a similar differential operator acting on the right, which will be denoted $\not\partial_\mu$. Using this notation, a different version of (3.1) becomes

$$df = \sum_\mu \chi_\mu \not\partial_\mu\, f.$$

There is an analogous expression for the right $\mathbb{H}$ action derived in appendix A. This formula also readily extends to quaternion valued forms, in which case the operator acts as an (anti-)derivation. For example we have

$$\mathrm{d}(\omega \wedge \nu) = (-1)^p \sum_\mu \left(\omega \not\partial_\mu \wedge \chi_\mu \wedge \nu + \omega \wedge \chi_\mu \wedge \not\partial_\mu \nu\right),$$

where ω is a p-form, with many possible variations from equations (A.4) and (A.5). It is also natural to define the conjugate Hodge-star operator, as used in complex differential geometry, which will be denoted $\bar{\star}$. Some useful formulas using this operator, proved in appendix B, are

$$\chi_\mu \wedge \bar{\star}\chi_\nu = -\bar{\star}\chi_\nu \wedge \chi_\mu = \delta_{\mu\nu}\, v, \quad \omega \wedge \bar{\star}\omega = <\omega, \omega> v, \tag{3.3}$$

where $<\omega, \omega>$ denotes the Hermitian inner product between forms, and v is the volume form. These formulas will be important when considering analogs of Cauchy's integral theorem.

To help interpret these operators and one forms, we can write a quaternion as an element of $\mathbb{C}^2$,

$$q = v + w\, \mathrm{e}_j \ : \ v = t + x\, \mathrm{e}_i, \quad w = y + z\, \mathrm{e}_i.$$

[1] There is some ambiguity in these definitions, for example we could choose $\bar{\not\partial}_\mu = \bar{\partial}_q\, \mathrm{e}_\mu$ and $\bar{\chi}_\mu \equiv \bar{\mathrm{e}}_\mu\, \mathrm{d}\bar{q}$, which simplifies the discussion of how rotations act on these operators. The definitions used here, however, provide significant simplifications when discussing higher degree forms, so we will use those conventions.

From twistor theory we know that this decomposition corresponds to a choice of embedding $\mathbb{H} \to \mathbb{C}^2$ that can be parameterized by a unit quaternion, which is the fiber of the mapping $P_3(\mathbb{C}) \to P_1(\mathbb{H})$ [3]

$$[v_1, w_1, v_2, w_2] \to [v_1 + w_1 j, v_2 + w_2 j],$$

corresponding to a choice of e_i. With this embedding, the $\boxslash$ operators become

$$\boxslash_0 = \partial_v - \mathrm{e}_j \partial_w, \quad \boxslash_i = \partial_v + \mathrm{e}_j \partial_w, \quad \boxslash_j = \partial_{\bar{v}} - \mathrm{e}_j \partial_{\bar{w}}, \quad \boxslash_k = \partial_{\bar{v}} + \mathrm{e}_j \partial_{\bar{w}}, \tag{3.4}$$

and the one-forms are similarly written

$$\chi_0 = \mathrm{d}v + \mathrm{d}w\,\mathrm{e}_j, \quad \chi_1 = \mathrm{d}v - \mathrm{d}w\,\mathrm{e}_j, \quad \chi_2 = \mathrm{d}\bar{v} + \mathrm{d}\bar{w}\,\mathrm{e}_j, \quad \chi_3 = \mathrm{d}\bar{v} - \mathrm{d}\bar{w}\,\mathrm{e}_j. \tag{3.5}$$

We can also investigate how these operators transform under rotation. For any proper rotation $\mathrm{d}x^R = R^T \mathrm{d}x$, there exist unit quaternions a and b such that $\mathrm{d}q^R = a\,\mathrm{d}q\,b$ and $\partial_q^R = \bar{b}\,\partial_q \bar{a}$. Under a reverse transformation, $\mathrm{e}'_\mu = \bar{a}\,e_\mu \bar{b}$, the one-forms transform as $\chi'^R_\mu = (a\mathrm{e}'_\mu b)\,\mathrm{d}q'^R\,(\bar{b}\,\bar{\mathrm{e}}'_\mu \bar{a})$, and similarly for $\boxslash_\mu$. In addition, summed combinations of the form $\mathrm{d} = \sum_\mu \chi_\mu \boxslash_\mu$ are rotationally invariant.

4 Quaternion integral formulas

It was shown by Fueter and Sudbery [8][9] that left regular functions, defined by $\bar{\boxslash}_0 f = 0$, satisfy an integral formula given by[2]

$$f(q_0) = \frac{1}{\pi^2} \int_{\partial\Omega} \frac{(q - q_0)^{-1}}{|q - q_0|^2} Dq\, f(q) \tag{4.1}$$

for $q_0 \in \Omega$. Here Dq is a quaternion valued three form, which turns out to be equal to $\bar{\star}\,\mathrm{d}\bar{q}$. Using the relations in (3.3), equation (4.1) is readily proved by using Stokes' theorem to find

$$\begin{aligned}
&\int_{\partial\Omega} \frac{(q - q_0)^{-1}}{|q - q_0|^2}\, \bar{\star}\,\mathrm{d}\bar{q}\, f(q) = \\
&\sum_\mu \int_\Omega \left[\, k(q - q_0) \boxslash_0 \bar{\star}\bar{\chi}_0 \wedge \bar{\chi}_\mu \bar{\boxslash}_\mu f(q) - k(q - q_0) \boxslash_0 \bar{\boxslash}_\mu\ \bar{\chi}_\mu \wedge \bar{\star}\bar{\chi}_0\, f(q)\right] = \\
&\qquad\qquad - \frac{1}{4} \int_\Omega \Delta\, k(q - q_0)\, \bar{\chi}_0 \wedge \bar{\star}\bar{\chi}_0\, f(q) = \pi^2 f(q_0),
\end{aligned} \tag{4.2}$$

where we have defined $k(q) = |q|^{-2}$. We can also define the general operator $\bar{\boxslash}_\eta = \eta\, \bar{\partial}_q\, \bar{\eta}$ for a unit quaternion η, with the corresponding one form, and consider functions f^η satisfying $\bar{\boxslash}_\eta f^\eta = 0$. A similar argument to (4.2) shows that these functions satisfy the integral formula

$$\pi^2 f^\eta(q_0) = \int_{\partial\Omega} k(q - q_0) \boxslash_\eta \bar{\star}\bar{\chi}_\eta\, f^\eta(q), \tag{4.3}$$

which suggests that f^η may be more naturally interpreted as a differential form. This point of view will be developed further.

To understand the structure of the differential forms, it is important to determine how antisymmetry of the wedge product translates to the basis χ_μ. To do this, it is useful to use the complex embedding defined in (3.5), while keeping e_i and e_j arbitrary until the end of the calculation. Using this embedding we find the equalities

$$\sum_\mu \chi_\mu \wedge \bar{\chi}_\mu = 0, \quad \chi_\mu \wedge \bar{\chi}_\nu = \chi_\nu \wedge \bar{\chi}_\mu, \quad \chi_0 \wedge \bar{\chi}_i = -\chi_j \wedge \bar{\chi}_k,$$

[2]Here Dq is half of the value in [9] to be consistent with our conventions.

with similar equations for the opposite ordering of χ and $\bar{\chi}$, which is related by a change of orientation. These equations are not nearly as simple if one considers expressions such as $\chi_\mu \wedge \chi_\nu$, probably because of their transformation under rotations. Additionally, we can interpret this basis for the two forms readily in terms of self-duality. In particular, we define the self-dual two-forms

$$\vartheta_i^+ = \mathrm{e}_i\, \bar{\chi}_0 \wedge \chi_i = -\frac{1}{2}(\mathrm{d}x_0 \wedge \mathrm{d}x_i + \mathrm{d}x_j \wedge \mathrm{d}x_k),$$
$$\omega_\mu^+ = \chi_\mu \wedge \bar{\chi}_\mu = \mathrm{e}_\mu \sum_i \mathrm{e}_i\, \vartheta_i^+\, \bar{\mathrm{e}}_\mu,$$

with a similar definition for the anti-self-dual forms (see appendix B). Some useful relations are

$$\omega_\mu^+ \wedge \omega_\nu^+ = -\frac{1}{2}(4\,\delta_{\mu\nu} - 1)\, v, \quad \omega_\mu^+ \wedge \chi_\nu = \frac{1}{2}\,(4\,\delta_{\mu\nu} - 1)\, \bar{\star}\bar{\chi}_\nu. \tag{4.4}$$

These equations provide all the tools necessary to evaluate three-forms and four-forms in terms of the basis we've defined. In appendix B we use this method to write the de Rham complex with this basis.

It is now possible to interpret (4.3) as an integral formula for differential forms. This is accomplished by using (4.4) to replace $\bar{\star}\bar{\chi}_\eta$ in equation (4.3), and rearranging terms. For example, if we consider a one-form $f^{(1)} = \sum_\mu \chi_\mu f_\mu^{(1)}$ satisfying $\bar{\not\partial}_\mu f_\mu^{(1)} = 0$, we find

$$\int_{\partial\Omega} \sum_\mu \Big(\chi_\mu(p) \otimes k(p-q) \,\bar{\wedge}_{\mu,q}\, \omega_\mu^+(q)\Big) \wedge f^{(1)}(q) = 2\pi^2 \Big(f^{(1)}(p) - \frac{1}{4}\sum_\mu \chi_\mu(p) \sum_\nu f_\nu^{(1)}(p)\Big).$$

The second term on the right prevents this from being a useful integral formula, unless we demand $\sum_\mu f_\mu^{(1)}(p) = 0$, which would make the term vanish. A different approach is to consider self-dual two forms, for which this condition comes from the fact that the ω_μ^+ form an over-complete basis. Indeed, for a self dual two-form $f^{(2)} = \sum_\mu \omega_\mu^+ f_\mu^{(2)}$ satisfying $\not\partial_\mu f_\mu^{(2)} = 0$, we can choose the coefficients $f_\mu^{(2)}$ to satisfy $\sum_\mu f_\mu^{(2)} = 0$, and we find

$$f^{(2)}(p) = \frac{1}{2\pi^2} \int_{\partial\Omega} G^+(p,q) \wedge f^{(2)}(q), \tag{4.5}$$

where we have defined

$$G^+(p,q) = \sum_\mu \omega_\mu^+(p)\, \bar{\not\partial}_\mu k(p-q) \otimes \bar{\chi}_\mu(q). \tag{4.6}$$

From (B.1) we find that this is an integral representation for a closed self-dual two-form. This formula leads directly to a generalization of equation (2.3) to the quaternions. To see this, we define the kernel taking values in $T^*\Omega \otimes T^*\Omega$

$$K(p,q) = \sum_\mu \frac{\chi_\mu(p) \otimes \bar{\chi}_\mu(q)}{|p-q|^2} = \sum_\mu \frac{\bar{\chi}_\mu(p) \otimes \chi_\mu(q)}{|p-q|^2} = 4\,\mathrm{Re}\left(\frac{\mathrm{d}p \otimes \mathrm{d}\bar{q}}{|p-q|^2}\right), \tag{4.7}$$

which satisfies the differential equation

$$-\,\mathrm{d}_p K(p,q) = G^+(p,q) + G^-(p,q),$$

where G^- is related to G^+ by a change of orientation. To prove this, we can insert d in the form $\sum_\mu \chi_\mu \not\partial_\mu$ into the first expression for K(p,q), and $\sum_\mu \bar{\chi}_\mu \bar{\not\partial}_\mu$ into the second expression. We then choose only the self-dual terms, and use the second form of the expression to get the anti-self-dual terms.

We use a similar method to find the second differential of K(p,q), as outlined in appendix C,

$$\mathrm{d}_p\mathrm{d}_q K(p,q) = \sum_i \Big[4\pi^2 \delta(p-q) \Big(\vartheta_i^+(p) \otimes \vartheta_i^+(q) \; + \; \vartheta_i^-(p) \otimes \vartheta_i^-(q) \Big) \\ + \sum_{j,k} \boxtimes_{jk} k(p-q) \Big(\vartheta_k^+(p) \otimes \vartheta_j^-(q) + \vartheta_j^-(p) \otimes \vartheta_k^+(q) \Big) \Big], \tag{4.8}$$

where we have defined the second order differential operators

$$\boxtimes_{jk} k(q) = 4\mathrm{Re}\Big(\partial_{\bar{q}}\, \mathrm{e}_j\, \partial_q\, \bar{\mathrm{e}}_k\, k(q) \Big) = \begin{cases} (\partial_0^2 + \partial_i^2 - \partial_j^2 - \partial_k^2)\, k(q) & \text{if } j = k, \\ (\epsilon_{ijk}\, \partial_{\{0,} \partial_{i\}} + \partial_{\{j,} \partial_{k\}})k(q) & \text{if } j \neq k. \end{cases} \tag{4.9}$$

These expressions lead to a generalization of (2.1) for the quaternions. For two closed self-dual two-forms f and g, we have

$$\int_\Omega f(p) \wedge g(p) = \frac{1}{2\pi^2} \int_{\partial\Omega} \int_{\partial\Omega} f(p) \wedge K(p,q) \wedge g(q), \tag{4.10}$$

In addition, if an arbitrary two form satisfies both $\mathrm{d}F = 0$ and $\mathrm{d}\bar{\star}F = 0$, then we obtain

$$\int_\Omega F(p) \wedge \bar{F}(p) = \frac{1}{2\pi^2} \int_{\partial\Omega} \int_{\partial\Omega} \mathrm{Re}\Big(F(p) \wedge K(p,q) \wedge \bar{\star} F(q) \Big) \tag{4.11}$$

This expression looks suggestively similar to a topological invariant that arises in the study of Yang-Mills theory, known in physics as the instanton number. This number represents an absolute lower bound for the energy of a field configuration in a given homotopy class, which is saturated by (anti-)self-dual connections known as instantons [3][21]. The conditions on F are also analogous to the Bianchi identity, and the Yang-Mills equation of motion. Our expression, however, is not currently relevant to Yang-Mills theory, because we are dealing with the exterior derivative instead of a gauge-covariant derivative. In the next section we will generalize these formulas to Yang-Mills theory.

5 Euclidean Yang-Mills theory

To begin our discussion of Yang-Mills theory, we will consider a Yang-Mills field A that takes values in $\mathrm{SU}(2) \otimes T^*M$, where SU(2) can also identified with the imaginary quaternions.[3] All of the manipulations performed will be valid for arbitrary Lie algebras, however, so we will use the trace instead of the real part, to keep the discussion general. The curvature of the field A is equal to

$$F = \mathrm{d}A + A \wedge A,$$

which satisfies the Bianchi identities

$$D_A F = \mathrm{d}F + A \wedge F - F \wedge A = 0. \tag{5.1}$$

In this paper, we will be considering solutions of the Yang-Mills equations of motion[4]

$$D_A \star F = 0. \tag{5.2}$$

[3] For a detailed description of formulating Euclidean SU(2) Yang-Mills theory with quaternions, see [3].

[4] Although this approach could potentially be used to consider arbitrary fields, the author has only found an expression for (a local form of) the instanton number, but not for the Lagrangian. The technical nature of this construction (involving a function introduced in [9]) does not seem to merit mentioning here, but may be useful in a different context.

which will be presented by adding and subtracting equations (5.1) and (5.2) to find

$$D_A F^+ = 0, \quad D_A F^- = 0. \tag{5.3}$$

We would like to derive a generalization of equation (4.10) to this case, where f(p)=g(p) becomes the (anti-)self-dual part of the curvature. For clarity we will consider the self-dual part, $F^+(p)$, for which we find

$$\mathcal{I}(F^+) \equiv \int_{\partial\Omega}\int_{\partial\Omega} \mathrm{Tr}\Big(F^+(p)\wedge K(p,q)\wedge F^+(q)\Big) = \int_{\Omega}\int_{\Omega} \mathrm{Tr}\Big(-\mathcal{A}(p,q)+\mathcal{B}(p,q)+\mathcal{C}(p,q)\Big),$$

where we have defined

$$\mathcal{A}(p,q) = \mathrm{d}_p F^+(p)\wedge K(p,q)\wedge \mathrm{d}_q F^+(q), \tag{5.4}$$

$$\mathcal{B}(p,q) = \mathrm{d}_p F^+(p)\wedge \mathrm{d}_q K(p,q)\wedge F^+(q) - F^+(p)\wedge \mathrm{d}_p K(p,q)\wedge \mathrm{d}_q F^+(q), \tag{5.5}$$

$$\mathcal{C}(p,q) = F^+(p)\wedge \mathrm{d}_p\mathrm{d}_q K(p,q)\wedge F^+(q). \tag{5.6}$$

We note that for our manifold $M \simeq \mathbb{R}^4$, we are considering integration over the product space $M\otimes M$, so forms from the different copies of M commute. In addition, in what follows, we will assume that the kernel K(p,q) takes symmetrized outer products, so that in this case K(p,q)=K(q,p). This, however, is for convenience, and the same results can be proved without this property.

To deal with the terms introduced by the Bianchi identities, we will first symmetrize the integrand with respect to p and q

$$\mathcal{I}(F^+) = \frac{1}{2}\int_{\partial\Omega}\int_{\partial\Omega} \mathrm{Tr}\Big(F^+(p)\wedge K(p,q)\wedge F^+(q) + F^+(q)\wedge K(q,p)\wedge F^+(p)\Big),$$

and then investigate how each term defined in (5.4) transforms under $(p\leftrightarrow q)$. For $\mathcal{A}(p,q)$, we use (5.3) to write

$$\mathcal{A}(p,q) = -\mathcal{A}_1(p,q) + \mathcal{A}_2(p,q)$$

where we have defined

$$\mathcal{A}_1(p,q) = \Big(A(p)\wedge F^+(p)\Big)\wedge K(p,q)\wedge\Big(F^+(q)\wedge A(q)\Big) + \Big(F^+(p)\wedge A(p)\Big)\wedge K(p,q)\wedge\Big(A(q)\wedge F^+(q)\Big),$$
$$\mathcal{A}_2(p,q) = \Big(A(p)\wedge F^+(p)\Big)\wedge K(p,q)\wedge\Big(A(q)\wedge F^+(q)\Big) + \Big(F^+(p)\wedge A(p)\Big)\wedge K(p,q)\wedge\Big(F^+(q)\wedge A(q)\Big).$$

$$\mathcal{B}_1(p,q) = \Big(A(p)\wedge F^+(p)\Big)\wedge \mathrm{d}_q K(p,q)\wedge F^+(q) - F^+(p)\wedge \mathrm{d}_p K(p,q)\wedge\Big(A(q)\wedge F^+(q)\Big),$$
$$\mathcal{B}_2(p,q) = \Big(F^+(p)\wedge A(p)\Big)\wedge \mathrm{d}_q K(p,q)\wedge F^+(q) - F^+(p)\wedge \mathrm{d}_p K(p,q)\wedge\Big(F^+(q)\wedge A(q)\Big).$$

$$\mathcal{C}(p,q) = F^+(p)\wedge \mathrm{d}_p\mathrm{d}_q K(p,q)\wedge F^+(q)$$

Using the cylic property and the symmetry of K(p,q), we find

$$\mathrm{Tr}\Big(\mathcal{A}_1(p,q)+\mathcal{A}_1(q,p)\Big)=0,\quad \mathrm{Tr}\Big(\mathcal{A}_2(p,q)+\mathcal{A}_2(q,p)\Big)=0,$$

where the first cancellation is inter-term and the second is intra-term. The inter-term cancellation for $\mathcal{B}(p,q)$ is similar, and is detailed in appendix D. It is easy to verify, however, that the terms do not vanish for $\mathcal{C}(p,q)$, which is the desired behavior.

The term-wise cancellation explained above allows us to extend the integral formula (4.10) quoted in section 4 to classical solutions of the Yang-Mills field equations, which is the central result of this paper. In particular, if we combine the self-dual and anti-self-dual parts, we find

$$\int_\Omega \mathrm{Tr}\Big(F(p)\wedge\star F(p)\Big) = \frac{1}{2\pi^2}\int_{\partial\Omega}\int_{\partial\Omega}\mathrm{Tr}\Big(F^+(p)\wedge K(p,q)\wedge F^+(q) - F^-(p)\wedge K(p,q)\wedge F^-(q)\Big),$$

which is a holographic expression for the action of a classical Euclidean Yang-Mills field on an arbitrary domain in $\mathbb{R}^4$.

A Conventions and Formulas

In this paper, we will not use the Einstein summation convention, because there will frequently be repeated indices that are not summed over. This is due to some of the definitions that are used, which simplify calculations overall. In addition, whenever i, j, and k appear as subscripts without summation, it will be assumed that $\mathrm{e}_i \mathrm{e}_j = \epsilon_{ijk} \mathrm{e}_k$.

For complex numbers, we will use the following convention to avoid factors of 2:

$$\mathrm{d}z = \mathrm{d}x + i\mathrm{d}y, \quad \partial_z = \frac{1}{2}(\partial_x - i\partial_y).$$

The Dolbeault operators are defined in the usual way, with the conventions following [22].

For the quaternions we will make a similar choice, defining

$$\mathrm{d}q = \frac{1}{2}\sum_\mu (\mathrm{e}_\mu \mathrm{d}x_\mu), \quad \partial_q = \frac{1}{2}\sum_\mu (\bar{\mathrm{e}}_\mu \partial_\mu).$$

We will also sometimes use the notation $\{t, x_i\}$ for the components of q.

The following formulas are used extensively throughout the text,

$$\mathrm{Re}(q) = \frac{1}{4}\sum_\mu \bar{\mathrm{e}}_\mu \, q \, \mathrm{e}_\mu, \quad \mathrm{Re}(pq) = \mathrm{Re}(qp). \tag{A.1}$$

Another set of useful relations, closely related to (A.1), are

$$q_\nu = \frac{1}{4}\sum_\mu \bar{\mathrm{e}}_\mu \, q \, \bar{\mathrm{e}}_\nu \mathrm{e}_\mu = \frac{1}{4}\sum_\mu \bar{\mathrm{e}}_\mu \bar{\mathrm{e}}_\nu \, q \, \mathrm{e}_\mu, \tag{A.2}$$

$$\bar{q} = -\frac{1}{2}\sum_\mu e_\mu \, q \, e_\mu. \tag{A.3}$$

We note that the the exterior derivative operator, $\mathrm{d} = \sum_\mu \mathrm{d}x_\mu \partial_\mu$, can be written $4\,\mathrm{Re}(\mathrm{d}q\,\partial_q)$, so using the definitions in equation (3.2), we write

$$\mathrm{d}f = \sum_\mu \chi_\mu (\boxslash_\mu f) = \sum_\mu (f \boxbslash_\mu)\, \chi_\mu, \tag{A.4}$$

$$\mathrm{d}f = \sum_\mu \bar{\chi}_\mu (\bar{\boxslash}_\mu f) = \sum_\mu (f \bar{\boxbslash}_\mu)\, \bar{\chi}_\mu. \tag{A.5}$$

Because all of these formulas come from writing the real (anti-commuting) exterior derivative with quaternions, we can immediately derive analogues of the product rule, such as

$$\mathrm{d}(\omega \wedge \nu) = (-1)^p \sum_\mu \left(\omega \, \bar{\boxbslash}_\mu \wedge \bar{\chi}_\mu \wedge \nu + \omega \wedge \chi_\mu \wedge \boxslash_\mu \nu \right).$$

It is also possible to write an arbitrary quaternion-valued form using a similar decomposition. In this case, we write $A = \sum_\mu a_\mu \mathrm{d}x_\mu$, and use equation (A.2) to write $\mathrm{d}x_\mu$ in terms of $\mathrm{d}q$, then rearrange terms to find $A = \sum_\mu \alpha_\mu \chi_\mu$, where $\alpha_\mu = \frac{1}{2}\sum_\nu (a_\mu \, \mathrm{e}_\mu \bar{\mathrm{e}}_\nu \bar{\mathrm{e}}_\mu)$. Conversely, we find that $a_\mu = \frac{1}{2}\sum_\nu (\alpha_\mu \, \mathrm{e}_\mu \mathrm{e}_\nu \bar{\mathrm{e}}_\mu)$. A similar procedure can be used for any degree form with any complete basis.

B Quaternion valued forms

A quaternion valued one-form is an element $\omega \in \mathbb{H} \otimes T^*M$ that acts on a quaternion valued vector $V \in \mathbb{H} \otimes TM$ by contraction and multiplication. The conjugate star operator acts naturally on quaternion valued one-forms when they are written using the χ_μ basis. In particular, the χ_μ basis satisfies

$$\chi_\mu \wedge \bar{\star}\chi_\nu = -\bar{\star}\chi_\nu \wedge \chi_\mu = \delta_{\mu\nu}\, v.$$

To prove this equation, we write $\chi_\mu \wedge \bar{\star}\chi_\nu = \mathrm{e}_\mu \mathrm{d}q\, (\mathrm{e}_\mu \bar{\mathrm{e}}_\nu) \wedge (\bar{\star}\mathrm{d}q)\, \bar{\mathrm{e}}_\nu$. Noting that only the middle terms are relevant, we calculate

$$\mathrm{d}q\, (\mathrm{e}_\mu \bar{\mathrm{e}}_\nu) \wedge \bar{\star}\mathrm{d}q = \frac{1}{24} \sum_{\alpha,\sigma} \mathrm{e}_\alpha (\mathrm{e}_\mu \bar{\mathrm{e}}_\nu) \bar{\mathrm{e}}_\sigma\, \epsilon_{\sigma\gamma\delta\rho}\, \mathrm{d}x^\alpha \wedge \mathrm{d}x^\gamma \wedge \mathrm{d}x^\delta \wedge \mathrm{d}x^\rho =$$
$$\frac{1}{4} \sum_{\alpha,\sigma} \mathrm{e}_\mu (\mathrm{e}_\mu \bar{\mathrm{e}}_\nu) \bar{\mathrm{e}}_\sigma\, \delta_{\alpha\sigma}\, v = \frac{1}{4}\, v \sum_\alpha \mathrm{e}_\alpha (\mathrm{e}_\mu \bar{\mathrm{e}}_\nu) \bar{\mathrm{e}}_\alpha = v\, \mathrm{Re}(e_\mu \bar{\mathrm{e}}_\nu) = v\, \delta_{\mu\nu}.$$

The second equality is proved in a similar way. This provides two natural bases for the three forms, given by $\bar{\star}\chi_\mu$ and $\bar{\star}\bar{\chi}_\mu$. The two forms are decomposed by direct calculation as described in the text, and using the techniques outlined we can write down the full de Rham complex in our basis:

$$\begin{aligned}
\mathrm{d}f^{(0)} &= \sum_\mu \chi_\mu \,\bar{\boxslash}_\mu\, f^{(0)}, \\
\mathrm{d} \sum_\mu \chi_\mu\, f^{(1)}_\mu &= -\sum_\mu \omega^+_\mu\, \bar{\boxslash}_\mu f^{(1)}_\mu - \frac{1}{2} \sum_i \omega^-_i \left(\bar{\boxslash}_{\{0} f^{(1)}_{i\}} - \bar{\boxslash}_{\{j} f^{(1)}_{k\}} \right), \\
\mathrm{d} \sum_\mu \left(\omega^+_\mu\, f^{(2)}_\mu + \omega^-_\mu\, g^{(2)}_\mu \right) &= 2 \sum_\mu \left(\bar{\star}\bar{\chi}_\mu \,\bar{\boxslash}_\mu\, f^{(2)}_\mu - \bar{\star}\chi_\mu \bar{\boxslash}_\mu g^{(2)}_\mu \right), \\
\mathrm{d} \sum_\mu \bar{\star}\chi_\mu f^{(3)}_\mu &= \sum_\mu \left(\bar{\boxslash}_\mu\, f^{(3)}_\mu \right) v,
\end{aligned} \tag{B.1}$$

where we have chosen $f^{(2)}_\mu$ to satisfy $\sum_\mu f^{(2)}_\mu = 0$, and a similar relation for $g^{(2)}$. Here, $\{\}$ represents symmetrization of indices.

It is also useful to have analogues of (4.4) for anti-self dual forms. We get

$$\omega^-_\mu \wedge \omega^-_\nu = \frac{1}{2} (4\, \delta_{\mu\nu} - 1)\, v, \quad \omega^-_\mu \wedge \bar{\chi}_\nu = -\frac{1}{2} (4\, \delta_{\mu\nu} - 1)\, \bar{\star}\chi_\nu.$$

The second identity is proved by writing the right hand side as an arbitrary linear combination of three forms, $\sum_\nu a_\nu \bar{\star}\chi_\nu$, then taking the wedge product of both sides with χ_ρ and comparing coefficients. A similar calculation shows that

$$\chi_\mu \wedge \vartheta^+_i = -\frac{1}{2}\, \mathrm{hs}(\mu_i)\, \bar{\star}\bar{\chi}_{\mu_i}\, \mathrm{e}_i, \qquad \bar{\chi}_\mu \wedge \vartheta^-_i = \frac{1}{2}\, \mathrm{hs}(\mu_i)\, \bar{\star}\chi_{\mu_i}\, \mathrm{e}_i, \tag{B.2}$$

where we have defined μ_i so that $\bar{\mathrm{e}}_\mu \mathrm{e}_{\mu_i} = \mathrm{qs}(\mu_i) \mathrm{e}_i$, and

$$\mathrm{hs}(\mu_i) = \begin{cases} 1 & \text{if } \mu = 0, i \\ -1 & \text{if } \mu = j, k, \end{cases} \qquad \mathrm{qs}(\mu_i) = \begin{cases} 1 & \text{if } \mu = 0, k \\ -1 & \text{if } \mu = i, j. \end{cases} \tag{B.3}$$

Some useful identities using these definitions are

$$\bar{\mathrm{e}}_i \mathrm{e}_{\mu_i} = \mathrm{hs}(\mu_i) \mathrm{qs}(\mu_i) \mathrm{e}_\mu, \quad \chi_\mu \wedge \bar{\chi}_{\mu_i} = -\mathrm{hs}(\mu_i) \vartheta^-_i \mathrm{e}_i. \tag{B.4}$$

It is also often necessary to take complex conjugates of the above listed equations, but we will not list these formulas here.

C Exterior derivative of integral kernel

The integral kernel K(p,q) was defined in equation (4.7), leading to expression (4.8) for the second differential. This expression is most easily obtained by using the definitions in (B.3) to get

$$\mathrm{d}_p\mathrm{d}_q \sum_\mu \chi_\mu(p)\, k(p-q)\, \bar\chi_\mu(q) = \sum_{\mu,\,\nu,\,\rho} (\chi_\mu \wedge \bar\chi_\nu(p) \bar\boxslash_\nu k(p-q) \boxslash_\rho \chi_\rho \wedge \bar\chi_\mu(q)) = \tag{C.1}$$

$$- \sum_{\mu,j,k} \mathrm{hs}(\mu_j) \Big(\mathrm{e}_j \bar\boxslash_{\mu_j} \boxslash_\mu \mathrm{e}_{k,\tilde\mu}\, k(p-q) \vartheta_j^-(p) \otimes \vartheta_k^+(q) + \mathrm{e}_{k,\tilde\mu} \bar\boxslash_\mu \boxslash_{\mu_j} \mathrm{e}_j\, k(p-q) \vartheta_k^+(p) \otimes \vartheta_j^-(q) \Big)$$

$$+ 4\pi^2 \delta(p-q) \sum_i (\vartheta_i^+(p) \otimes \vartheta_i^+(q) + \vartheta_i^-(p) \otimes \vartheta_i^-(q)),$$

where $\mathrm{e}_{i,\tilde\mu} = \mathrm{e}_\mu \mathrm{e}_i \bar{\mathrm{e}}_\mu$. Use of the identities in (B.4) then leads directly to equation (4.8).

D Cancellation of B(p,q) terms

To show that the $\mathcal{B}(p,q)$ term vanishes when $(p \leftrightarrow q)$, we write

$$\mathcal{B}(p,q) = \mathcal{B}_1(p,q) + \mathcal{B}_2(p,q)$$

where we have defined

We can then show that

$$\mathrm{Tr}\Big(\mathcal{B}_1(p,q) + \mathcal{B}_1(q,p)\Big) = 0, \quad \mathrm{Tr}\Big(\mathcal{B}_2(p,q) + \mathcal{B}_2(q,p)\Big) = 0,$$

which is the desired cancellation.

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